BJOG: an International Journal of Obstetrics and Gynaecology
June 2003, Vol. 110, pp. 593–597

The Doppler Assessment in multiple pregnancy randomised controlled trial of ultrasound biometry *versus* umbilical artery Doppler ultrasound and biometry in twin pregnancy

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Names of group collaborators may be found on page 596

Objective To assess the addition value of umbilical artery Doppler ultrasound added to standard ultrasound biometry measurements in the management of twin pregnancies.

Design A prospective randomised controlled multicentre trial of women with twin pregnancies.

Setting Tertiary level referral hospitals in Australia, New Zealand and Southeast Asia.

Population Pregnant women with twin pregnancies.

Methods Women were randomised at 25 weeks of gestation to receive either standard ultrasound biometric assessment or standard assessment plus Doppler ultrasound umbilical artery flow velocity waveform analysis. The studies were repeated at 30 and 35 weeks unless otherwise indicated. Physicians were advised to institute close fetal surveillance in the presence of an abnormal umbilical artery Doppler study or with biometry indicators of fetal compromise.

Main outcome measures Standard obstetric (mode of delivery, perinatal mortality, hypertension, antenatal admissions and gestation at delivery) and neonatal (5 minute Apgar scores <5, admissions to neonatal nursery and requirements for ventilation) outcomes and statistical analysis was on intention-to-treat.

Results There were no significant differences between the two groups with respect to demography, antenatal, peripartum and neonatal outcomes. There was no difference in the perinatal mortality rate in the no Doppler group ($n = 264$), which was 11/1000 live births, and the Doppler group ($n = 262$), which was 9/1000 live births. There were three unexplained intrauterine deaths in the no Doppler group and none in the Doppler group (OR 0.14, 95% CI 0.01–1.31). Two intrauterine deaths in the Doppler group were due to cord prolapse in labour and a fetomaternal haemorrhage, both very unlikely to be influenced by Doppler surveillance.

Conclusions In this study, close surveillance in twin pregnancy resulted in a lower than expected fetal mortality from 25 weeks of gestation in both the no Doppler and Doppler groups. The lower rate of unexplained fetal death in the no Doppler group was not significantly different from the Doppler group.

INTRODUCTION

The use of Doppler ultrasound in the evaluation of umbilical artery velocity waveforms in high risk pregnancies has been exhaustively assessed in randomised controlled trials[1,2]. Neilson and Alfirevic[3] have reported that Doppler ultrasound in high risk pregnancies has been associated with significant reductions in perinatal mortality, the rate of stillbirths in normally formed infants, antepartum admission to hospital, induction of labour, elective delivery and caesarean section for fetal distress with an increase in deliveries <34 weeks of gestation. High risk pregnancy status was quite heterogeneous and has been variously described as 'high risk', those referred for fetal assessment, those who aroused concern regarding fetal wellbeing, those at risk of a small for gestational age baby, where the abdominal circumference was <5th centile on ultrasound, those women admitted to the antenatal ward and those with hypertension or a small for gestational age fetus >28 weeks of gestation. Neilson and Alfirevic[3] have suggested that the evaluation of Doppler should be extended to specific high risk groups of pregnancies. One such group of pregnancies is that of

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doi:10.1016/S1470-0328(03)02966-5

twin pregnancies. Adequate data for assessment of Doppler in the management of twin pregnancies have only been available from two small randomised controlled trials[4,5]. These two trials gave a combined odds ratio (OR) of 0.14 (95% confidence intervals [CI] 0.03–0.77) for a reduction in intrauterine death. However, the groups of twin pregnancies studied were 48 control pregnancies and 34 pregnancies in which Doppler was used in the management.

There are two reported historically controlled studies that have shown decreases in perinatal mortality of 79%[6] and 60%[7] with a 16% reduction in admissions to neonatal intensive care[6]. In contrast to these two reports, there have been a variety of studies either supporting[8–15] or not supporting[16–20] the use of Doppler ultrasound in twin pregnancies. It is very likely that the reasons for these widely variable reports of the use of Doppler were due to the variable criteria for the assessment of adverse outcomes, varying from the presence of fetal growth restriction to perinatal mortality.

In view of the limitations of the observational studies and the previous historically controlled studies, we undertook this randomised controlled trial to assess the impact of the addition of umbilical artery Doppler waveform assessment to the current ultrasound biometry assessment of twin pregnancies.

METHODS

To enable recruitment, sufficient number of twins collaboration was sought with several Australian and Southeast Asian maternal fetal medicine units. The Australian units were John Hunter Hospital (Newcastle), Mater Mother's Hospital (Brisbane), Royal Women's Hospital (Melbourne), Canberra Hospital (Australian Capital Territory) and the Mother's and Children's Hospital (Adelaide). The international hospitals were National Women's Hospital (New Zealand), Wellington Hospital (New Zealand), KK Women's and Children's Hospital (Singapore) and the Department of Obstetrics and Gynaecology Kuala Lumpur (Malaysia). The approval to conduct the study was obtained from each institution's research and ethics committee. The recruitment of patients was at 25 weeks of gestation when the presence of two viable apparently normally formed twins was seen on ultrasound. Twin pregnancies that were complicated by fetal anomalies, polyhydramnios/oligohydramnios sequence or demise of one twin before 25 weeks were excluded from the study. The complication of polyhydramnios/oligohydramnios sequence was excluded on the grounds that where this condition has commenced prior to 25 weeks of gestation there are regimens for treatment specific for this condition that do not rely on Doppler ultrasound. As this was to be a pragmatic study to be undertaken in a large number of units, no attempt was made to classify the twins by their apparent chorionicity as the significance of determination of chorionicity was not realised at that time the study was commenced.

After informed consent was obtained, randomisation was undertaken using opaque sealed envelopes containing the randomisation code, the envelope being opened by an observer remote from the patient's care. The envelopes were issued from John Hunter Hospital in blocks of 20 to the recruiting institutions.

The recruitment and randomisation protocol is shown in Fig. 1. Ultrasound assessment of the women was undertaken at randomisation at 25 weeks of gestation and then at 30 and 35 weeks of gestation unless otherwise indicated. The clinicians were advised to undertake interventions if there was an abnormal umbilical artery Doppler study (>95th centile systolic diastolic ratio)[21] or abnormal ultrasound biometry indicating discordant growth[22]. The suggested intervention was intensive surveillance by the obstetric caregivers (i.e. non-stress cardiotocography, ultrasound growth assessment and amniotic fluid volume assessment). If other indicators of fetal wellbeing (lack of serial growth, decreased amniotic fluid volume or abnormal fetal heart rate monitoring) were abnormal, then early delivery was advised from the 25th week study at the discretion of the caregiver. An abnormality of the Doppler waveforms themselves was not considered an indication for immediate delivery unless there was absence of diastolic flow velocity at >32 weeks of gestation.

Maternal data collected were age, previous obstetric history, antenatal admissions, presence of hypertension, gestation at delivery, indication for delivery and mode of delivery. Fetal data were ultrasound biometry measurements, umbilical artery Doppler systolic diastolic ratios and the occurrence of fetal death and causative factors. Neonatal data were birthweights, Apgar scores, admission

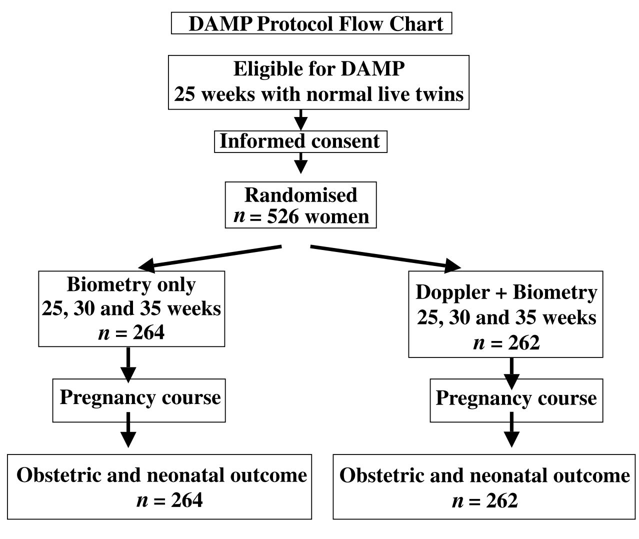

Fig. 1. Flow chart of DAMP randomisation protocol.

Table 1. Demographic data for the no Doppler ($n = 264$) and Doppler ($n = 262$) groups of twins. Values are given as mean [SD] and median (interquartile range).

	Doppler ($n = 262$)	No Doppler ($n = 264$)
Maternal age	29.3 [5.4]	29.3 [5.6]
Previous live births	0 (0–1.75)	1.0 (0–2)
Previous stillbirths	0 (0)	0 (0)
Previous <20 weeks loss	0 (0–1)	0 (0–1)
Antenatal admissions (days)	0 (0–3)	0 (0–3)
Delivery gestation (weeks)	35.8 [2.8]	35.7 [2.9]

to neonatal intensive care, admission to special care nursery, requirements for ventilation and occurrence of neonatal death (up to 28 days of life).

In calculating the sample size for the proposed Doppler Assessment in Multiple Pregnancy (DAMP) study, the perinatal mortality for twins in the Hunter Region of New South Wales Australia was considered for the year 1990. In that year, there were 70 twin pregnancies delivered with a perinatal mortality of normally formed babies after 25 weeks of 6 (i.e. 8.6%, or 85.7/1000 live births). A reduction in perinatal mortality by 50% from 80/1000 to 40/1000 was considered to be a clinically significant primary outcome hypothesis. The number of women with a twin pregnancy required to be randomised into each arm of the trial was calculated by nQuery Adviser, release 3.0 (Statistical Solutions, Corke, Ireland) and was 310. This took into account the clustered nature of the data with respect to the fact that a significant clinical event affecting one of the twin pair would impact upon the co-twin. This number gave an alpha value of 0.05 with power of 0.80 to detect a significant reduction in perinatal mortality using a two-tailed test at the 5% significance level[23]. Secondary hypotheses studied were that there would be reductions in fetal deaths, antenatal admissions to hospital, induction of labour and emergency caesarean sections (i.e. caesarean sections in labour).

Statistical analysis was taken on an intention-to-treat analysis. The analysis was performed with Minitab release 12.22 (Minitab, 1998, Pasadena, California) with non-parametric testing for continuous variables and assessment of OR with 95% CI for categorical outcomes.

RESULTS

There were 539 women randomised in the DAMP study to either standard ultrasound biometry with ($n = 268$) or without umbilical artery Doppler Waveform studies ($n = 271$). Of these, 13 women were lost to follow up after randomisation at 25 weeks and were not able to be included in the results. Following randomisation, the women at 25 weeks of gestation were studied again at 30 and 35 weeks of gestation. This left 526 with complete follow up for analysis (262 in the Doppler group and 264 in the no Doppler group. Assessment of the maternal demographic factors (Table 1) shows that the randomisation process was successful. The study protocol was followed except for seven cases in the no Doppler group who had Doppler ultrasound performed. There were no significant differences in the antenatal, intrapartum or neonatal outcomes for the two groups (Table 2).

The patients were randomised for this study from March 1993 up to and including March 1997. At this point, it was apparent that the adverse outcome of perinatal mortality rate in the no Doppler group was 11/1000 live births, much lower than the originally calculated rate of adverse outcome. The original sample size of 310 per arm of the trial was therefore unable to detect a halving of the perinatal mortality. To detect such a difference, 3300 per arm would have been needed.

There were six neonatal deaths, three in each group (one neonate dying at 55 days of age after delivery at 25 weeks of gestation where the co-twin died at 22 days of age). All were delivered at <27 weeks of gestation.

Table 2. Outcome variables for the no Doppler and Doppler groups of twin with relative risks (RR) and 95% CI. Values are given as n and RR (95% CI).

	Doppler ($n = 262$)	No Doppler ($n = 264$)	RR (95% CI)
Antenatal outcomes			
Diagnosis of hypertension	52	59	0.86 (0.57, 1.31)
Antenatal admission	120	126	0.93 (0.66, 1.30)
Delivery outcomes			
IOL	62	57	1.13 (0.75, 1.69)
IOL and vaginal delivery	53	50	1.09 (0.71, 1.67)
Spontaneous vaginal delivery	157	153	1.08 (0.77, 1.53)
Elective LSCS	58	55	1.08 (0.71, 1.64)
Emergency LSCS	40	48	0.81 (0.51, 1.28)
Emergency CS twin 2	9	5	1.81 (0.63, 5.23)
Perinatal outcomes			
All cases of IUFD	2	3	0.67 (0.12, 3.91)
Unexplained IUFD	0	3	0.14 (0.01, 1.31)
5 minute Apgar <5 twin 1	2	2	1.00 (0.14, 7.14)
5 minute Apgar <5 twin 2	2	4	0.51 (0.1, 2.57)
Neonatal deaths to 28 days	5	5	1.01 (0.29, 3.52)
Perinatal mortality to 28 days	7	8	0.88 (0.32, 2.45)
Neonatal outcomes			
NICU admission twin 1	37	34	1.11 (0.67, 1.83)
NICU admission twin 2	39	42	0.92 (0.58, 1.48)
Ventilation twin 1	22	25	0.88 (0.48, 1.60)
Ventilation twin 2	25	30	0.82 (0.47, 1.44)
SCN admission twin 1	122	133	0.86 (0.61, 1.21)
SCN admission twin 2	131	136	0.94 (0.67, 1.32)

IOL = induction of labor; LSCS = lower segment caesarean section; CS = caesarean section; IUFD = intrauterine fetal death; NICU = neonatal intensive care unit; SCN = special care nursery.

There were three fetal deaths in the no Doppler group. All three occurred in the third trimester and known causes of fetal demise were excluded (namely, fetomaternal haemorrhage, twin–twin transfusion and cord accident). All three had normal ultrasound biometry at the 30-week study and on that occasion there were no indications of fetal compromise. Following delivery, two of the three were noted to have monochorionic placentation and the third had dichorionic placentation. The two fetal deaths that occurred in the Doppler group were one case of cord prolapse in labour at 36 weeks of gestation and one case of a 77 mL fetomaternal haemorrhage at 36 weeks of gestation in a dichorionic twin pair. We do not believe that Doppler ultrasound could be expected to change these two outcomes.

DISCUSSION

The results from this randomised assessment of the addition of Doppler ultrasound of the fetal umbilical arterial circulation in twin fetuses has fulfilled the suggestion of Alfirevic and Neilson[2] and Neilson *et al.*[17] that specific high risk pregnancy groups be assessed for the role of Doppler ultrasound[3]. In the historical context of the decade between the planning and conception of this study and this report, there has been an appreciable improvement in the overall perinatal mortality in twin pregnancies. This is clearly seen if one considers the background rates of perinatal death in twins in the observational and randomised controlled studies[5] of that era. There is clearly no doubt that increased awareness of clinicians of the difficulties of being a twin fetus has resulted in the improvements seen over this period. In our study, twin fetuses in the no Doppler group had a perinatal mortality rate of 11.4/1000 live births. However, this increased awareness and improvement did not prevent the three unexplained intrauterine fetal deaths in the no Doppler group. The two fetal deaths in the Doppler group were due to unforeseen complications in the form of cord prolapse in labour and fetomaternal haemorrhage, both of which were very unlikely to be predicted from the Doppler studies.

Although this study did not address the issue of chorionicity, this is undoubtedly an important issue for the current management of twins. Two of the three unexplained fetal deaths *in utero* in our study had monochorionic placentation. When this study was planned, the ability of correct diagnosis of chorionicity in the first trimester with ultrasound was not widely realised[24]. Now that the close association between chorionicity and poorer fetal outcome is recognised, the importance of establishing chorionicity early in pregnancy must be stressed as this is likely to be the group of twins that will be best served by Doppler surveillance as they form the highest risk group of twins.

DAMP Group Collaborators

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Accepted 28 January 2003